

FACMIC

FEDERATED ADAPTATIVE CLIP MODEL FOR MEDICAL IMAGE CLASSIFICATION

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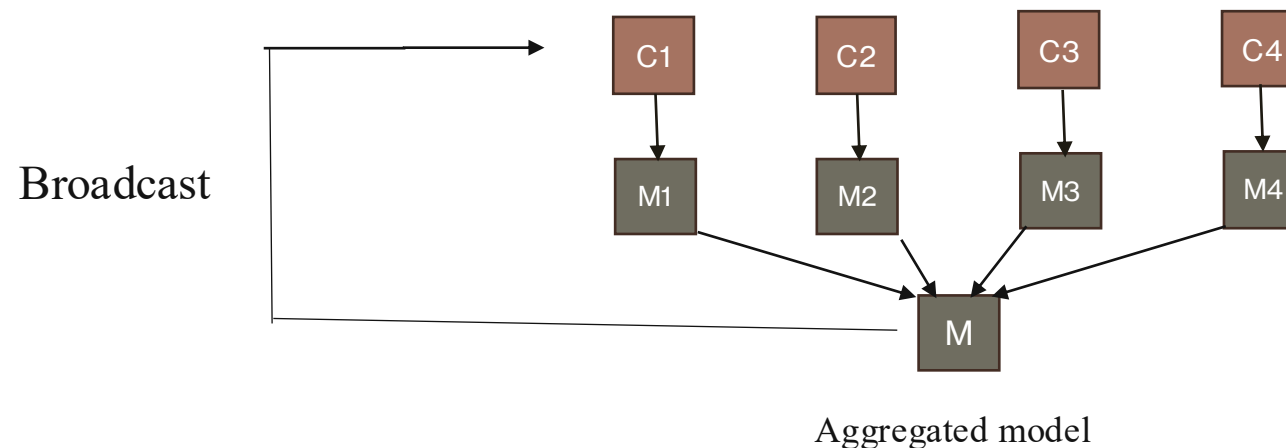
MOTIVATION

Deep Learning :

- Modelling paradigm that uses deep neural networks
- Requires large datasets
- Assumes centralized data access

Federated Learning :

- Federated Learning came into light after recent growing focus on data security and privacy.
- Assumes decentralized data access.
- FL enables model aggregation without directly accessing the raw user data from different clients.



MOTIVATION

Factors influencing FL :

- Communication cost in FL
- Data distribution shifts

Why foundational models are not very useful-

- Vision Language Models (VLMs) like CLIP are extremely large (with over 10^8 parameters)-This would incur huge communication cost-Transferring CLIP weights is prohibitively expensive

But we need a FL framework that-

- | | |
|---------------------------------------|--|
| ⌚ Preserves data privacy | ⌚ Can adapt to different data distributions |
| ⌚ Uses foundational model efficiently | ⌚ Will work for real world multi-source medical data |
| ⌚ Reduces communication cost | |

PROBLEM STATEMENT

Goal: Classifying images to their corresponding classes using image-text pair.

FACMIC Setting:

- We consider N clients (e.g. Hospitals) namely $C_1, C_2, \dots, C_N$
- Each client holds a private medical image dataset – i -th client C_i has dataset $D_i = \{(x_{i,j}, y_{i,j})\}, j=1, 2, \dots, n_i$
- Data cannot be shared among clients.
- We assume that all clients have same input and output space but the data distribution are different for different clients i.e. $P(D_i) \neq P(D_{i'})$ for all $i \neq i'$
- Each dataset D_i consists of three non-overlapping subsets- Training set (D_i^{train}), Validation set (D_i^{val}) & Test set (D_i^{test})
- We want to train a robust global model $f_\theta(\cdot)$ that preserves data privacy of each client
- This model should perform well on test data of each of the clients i.e.

$$\min_f \frac{1}{N} \sum_{i=1}^N \frac{1}{n_i^{test}} \sum_{j=1}^{n_i^{test}} l(f_\theta(x_{i,j}^{test}), y_{i,j}^{test})$$

- Also, this model should perform well on any client's data which were excluded from training stage.

CHALLENGES

There are three core constraints that our solution must satisfy-

1. Privacy constraint-
 - No centralized training allowed
2. Communication cost constraint-
 - Parameter-heavy foundational models like CLIP are too large to share
3. Data distribution heterogeneity-
 - Client data distributions differ due to –
 - a. Imaging devices
 - b. Acquisition protocols
 - Distribution discrepancy across clients' impact FL prediction accuracy negatively

EXISTING WORKS WITH LIMITATIONS

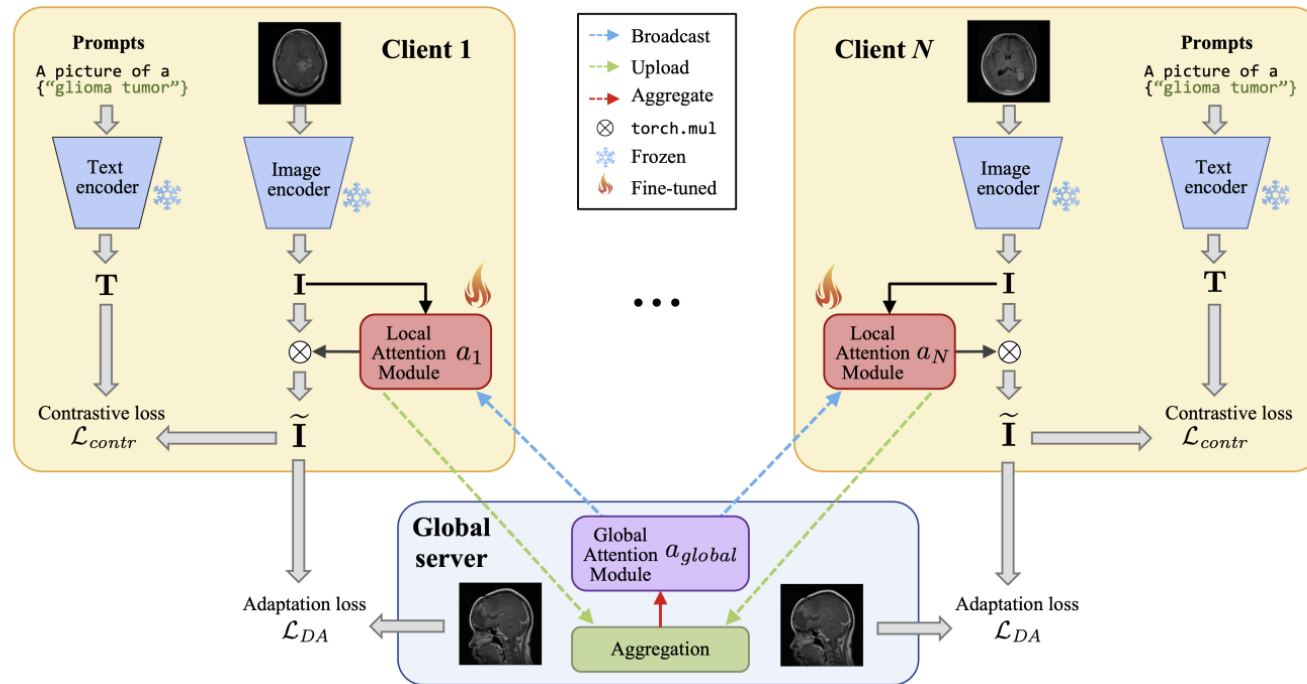
Work on FL with foundational models have started recently, below are some existing works-

- Federated Learning for healthcare applications :
Introduced FL to various medical datasets , also hinted about problems FL faces due to data distribution discrepancies.
- Robust and communication-efficient federated learning from non-iid data : shows that the performance of FL models is typically influenced by two main factors: data distribution shifts and communication costs.
- On the opportunities and risks of foundation models : shows why it is problematic to use of large Vision Language Models (VLMs) such as CLIP with large number of parameters.
- Are natural domain foundation models useful for medical image classification? : shows that foundation models like CLIP fail for medical image classification tasks (e.g., 52.7% classification recall for CLIP using the ISIC2019 dataset)
- BiomedCLIP: BiomedCLIP introduces a large-scale biomedical vision–language foundation model trained on 15 million image–caption pairs from PubMed Central. State of the Art performance in biomedical retrieval, classification but no application yet in the FL setup.
- FedCLIP: Introduced generalization and personalization for CLIP models using an adapter module in the Image encoder of the CLIP, but does not address the domain alignment issue.

METHODOLOGY

FACMIC Framework

Federated Adaptative CLIP Model for Medical Image Classification



Uses a pre-trained CLIP model with an image encoder & text encoder

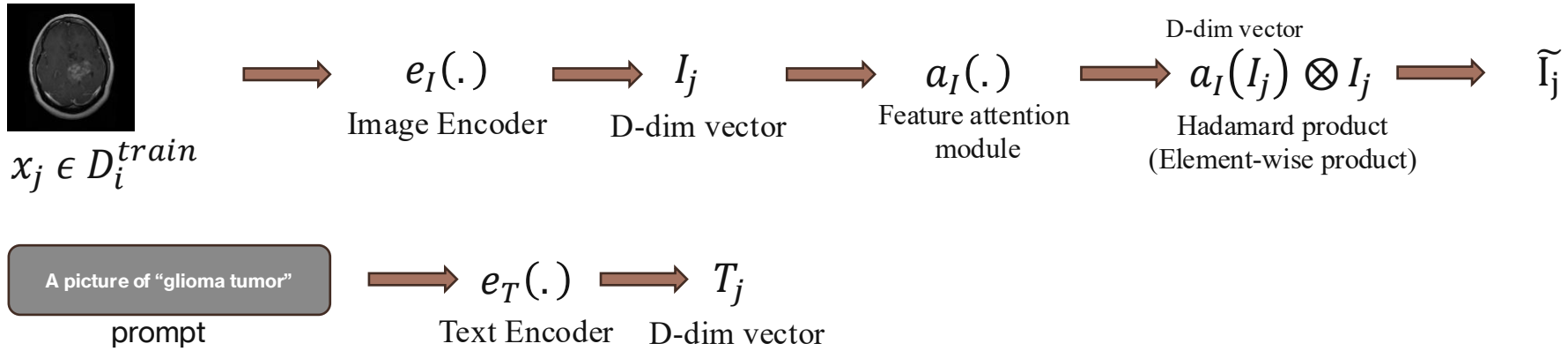
FACMIC adds a light-weight feature attention module on top of CLIP model

Also introduces a domain adaptation (DA) strategy to minimize data discrepancies between clients

Image and text encoders are frozen- only the attention module is trained

METHODOLOGY

FACMIC Framework



We measure the probability of x_j belongs to a class c by measuring cosine similarity of $\tilde{I}_j$ and the text features T_c corresponding to the prompt of class c by,

$$P(Y = c / x_j) = \frac{e^{(s_{j,c}/\tau)}}{\sum_{c'=1}^K e^{(s_{j,c'}/\tau)}}$$

where,

$$s_{j,c} = \frac{\langle \tilde{I}_j, T_c \rangle}{\| \tilde{I}_j \| \cdot \| T_c \|}$$

here τ is the softmax temperature parameter that controls how sharp or smooth the softmax output distribution is.

METHODOLOGY

FACMIC Framework

To train our model we have to define the loss function. The loss function consists of two parts-

- Contrastive loss ($\mathcal{L}_{\text{contr}}$)
- Domain adaptation loss ($\mathcal{L}_{\text{DA}}$)

Contrastive Loss

- Pushes together the image and text features from the same training example and pulls apart non-matching ones
- $\mathcal{L}_{\text{contr}}$ is computed over batches of size B
- A matrix S is defined which is of size $B \times B$. $S_{j,j'} = \text{cosine similarity of the features } \tilde{I}_j \text{ and } T_{j'}$
- An image probability matrix P is defined $P = \text{softmax}(S / \tau) \in [0,1]^{B \times B}$
- A text probability matrix Q is defined as $Q = \text{softmax}(S^T / \tau) \in [0,1]^{B \times B}$
- By the definition of contrastive loss, we want, for each pair j
 - image $j \longrightarrow \text{text } j$
 - text $j \longrightarrow \text{image } j$

therefore , we only care about diagonal probabilities. For one pair j , the loss is $\ell_j = -\frac{1}{2} (\log P_{j,j} + \log Q_{j,j})$

- For the entire batch,

$$\mathcal{L}_{\text{contr}} = -\frac{1}{2B} \sum_{j=1}^B (\log P_{j,j} + \log Q_{j,j})$$

- Batch-wise contrastive loss makes each image identify its own text among all others in the batch—and vice versa—by sharpening the diagonal of the similarity matrix.

METHODOLOGY

FACMIC Framework

Contrastive Loss (Example)

Suppose $B=4$. So, there are 4 image-text pairs in one batch. Say – $[(I_1, T_1), (I_2, T_2), (I_3, T_3), (I_4, T_4)]$. Here I_j, T_j are both D -dimensional vectors.

Consider the similarity matrix S is the following -

	T1	T2	T3	T4
I1	0.92	0.11	0.05	0.02
I2	0.10	0.88	0.20	0.15
I3	0.03	0.25	0.91	0.10
I4	0.08	0.12	0.18	0.95

If we compute P and Q matrix using softmax function , we get-

$$P = \begin{bmatrix} 0.440 & 0.196 & 0.184 & 0.179 \\ 0.187 & 0.409 & 0.207 & 0.197 \\ 0.174 & 0.217 & 0.421 & 0.187 \\ 0.181 & 0.188 & 0.200 & 0.431 \end{bmatrix}$$

$$Q = \begin{bmatrix} 0.438 & 0.188 & 0.177 & 0.174 \\ 0.193 & 0.406 & 0.205 & 0.198 \\ 0.180 & 0.217 & 0.418 & 0.188 \\ 0.189 & 0.190 & 0.201 & 0.440 \end{bmatrix}$$

While minimizing the contrastive loss , we basically want to increase the principal diagonal elements of P and Q and suppress the off diagonal elements so that ,the parameters are updated in such a manner that the model is able to identify correct image-text pairs.

METHODOLOGY

FACMIC Framework

Domain Adaptation Loss

FACMIC introduces a client-specific feature adaptation strategy based on the Local Maximum Mean Discrepancy (LMMD) method

- The assumption is that this data distribution shift occurs only on image features.
- The loss function is defined as-

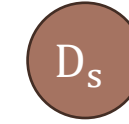
$$\mathcal{L}_{DA} = \frac{1}{K} \sum_{c=1}^K \left\| \sum_{\mathbf{x}_i^s \in \mathcal{D}_s} \omega_{i,c}^s \phi(\mathbf{x}_i^s) - \sum_{\mathbf{x}_j^t \in \mathcal{D}_t} \omega_{j,c}^t \phi(\mathbf{x}_j^t) \right\|_{\mathcal{H}}^2, \quad \omega_{i,c}^{s|t} = \frac{\mathbb{1}(y_i = c)}{\sum_{y_j \in \mathcal{D}^{s|t}} \mathbb{1}(y_j = c)}. \quad (5)$$

Here, $\phi(\cdot)$ is a mapping function to a Hilbert space $\mathcal{H}$, while $\omega_{i,c}^s, \omega_{j,c}^t$ are weights that measure the membership of example $\mathbf{x}_i^s$ and $\mathbf{x}_j^t$ in class c .

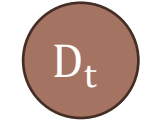
Since we can not compute $\phi(\cdot)$ directly, we use the kernel trick and the loss becomes-

$$\mathcal{L}_{DA} = \frac{1}{K} \sum_{c=1}^K \left[\sum_{i=1}^{n_s} \sum_{j=1}^{n_s} \omega_{i,c}^s \omega_{j,c}^s k(\mathbf{I}_i^s, \mathbf{I}_j^s) + \sum_{i=1}^{n_t} \sum_{j=1}^{n_t} \omega_{i,c}^t \omega_{j,c}^t k(\mathbf{I}_i^t, \mathbf{I}_j^t) - 2 \sum_{i=1}^{n_s} \sum_{j=1}^{n_t} \omega_{i,c}^s \omega_{j,c}^t k(\mathbf{I}_i^s, \mathbf{I}_j^t) \right] \quad (6)$$

where n_s, n_t are the number of samples in the source domain and target domain, respectively, and $k(\cdot, \cdot) = \langle \phi(\cdot), \phi(\cdot) \rangle$ is the kernel function.



Source Domain
(Client Data)



Target Domain
(Public data)

Compare the class-wise distribution statistics between D_s and D_t using the LMMD method

But public data is unlabeled. We compute Pseudo-labels by the following formula we have used before-

$$P(Y = c / \mathbf{x}_j) = \frac{e^{(s_{j,c}/\tau)}}{\sum_{c'=1}^K e^{(s_{j,c'}/\tau)}}$$

We assign the image to the class with highest Probability and then proceed with the calculation of

$\mathcal{L}_{DA}$

METHODOLOGY

FACMIC Framework

Final loss function

$$\mathcal{L} = \mathcal{L}_{\text{contr}} + \lambda \mathcal{L}_{\text{DA}}$$

Where λ is a hyper-parameter controlling the trade-off between these two losses

Global aggregation

- aggregation strategy to combine the parameters of different clients into a single global model.
- In each round, each client C_i uploads its parameters θ_i^a to the global server, then the global server aggregates these parameters into a single vector θ_{global}^a using a weighted average –

$$\theta_{\text{global}}^a = \sum_{i=1}^N \omega_i \cdot \theta_i^a, \quad \omega_i = \frac{n_i^{\text{train}}}{\sum_{i'=1}^N n_{i'}^{\text{train}}}.$$

- Next, the server broadcasts the global attention module parameters back to each client
- Since, the introduced attention module has very few parameters it is easily trainable – with a low communication and computation cost

IMPLEMENTATION

- The ViT-B/16 (Vision Transformer Base / Input images are divided into 16 x 16 patches to capture context better) is used as the backbone CLIP model
- The feature attention module consists of 5 layers- a linear layer, a batch normalization layer, a LeakyRelu layer, a second linear layer and final softmax activation layer.
- For the LMMD loss , a Gaussian kernel is used for the kernel trick.
- λ is set to 1.
- During training CLIP encoders are frozen and only attention module is trained
- Adam optimizer is used with $\beta_1 = 0.9$ and $\beta_2 = 0.98$, weight decay of 0.02, fixed learning rate = 5×10^{-5} & batch size is 32
- As data pre-processing, we resized the image to 224 x 224 and normalized their intensity using z-score normalization for both the training and testing phases.
- The environment used for experiments is based on the Windows 11 operating system, and features an Intel 13900KF CPU with 128 GB of RAM and an RTX 4090 GPU.

EXPERIMENTS



DATASET

BT (Brain Tumor)

- ❖ 4 classes
- ❖ 2870 training samples
- ❖ 394 testing samples
- ❖ 100 training rounds

SC (Skin cancer)

- ❖ Obtained from ISIC
- ❖ 9 classes
- ❖ 2239 training samples
- ❖ 118 testing samples
- ❖ 50 training rounds

Real multi-source skin cancer

- ❖ ISIC2019+ SC+HAM10000
- ❖ 7 classes
- ❖ 30000 training samples
- ❖ 6233 testing samples
- ❖ 50 training rounds



CLIENT DIVISION

- **IID distribution**-for BT,SC Divide training samples into 3 equal sizes as 3 clients
- **Non-IID distribution** – for BT,SC use Dirichlet distribution, with concentration parameter $\alpha = \{0.3, 0.6, 0.9\}$ to generate non-iid distribution among clients-

Samples of each client under non-iid conditions for BT and SC datasets.

Clients	$\alpha = 0.3$		$\alpha = 0.6$		$\alpha = 0.9$	
	BT	SC	BT	SC	BT	SC
Client1	2075	218	1355	387	1480	1156
Client2	389	1374	908	487	423	335
Client3	406	647	607	1365	967	748
Global	394	118	394	118	394	118

- **For the Real dataset**- Client 1 (SC) has 1971 samples, Client 2 (ISIC2019) has 19766 samples, Client 3 (HAM10000) has 8512 samples. Different sources act as different clients

RESULTS

- ❖ Authors compare the results of FACMIC with state-of-the-art methods – FedAVG, MOON, FedProx, FedFocal, FedCLIP & a centralized method.
- ❖ Accuracy metric (ACC) is used for all the datasets. In the real dataset due to class imbalance Balanced Accuracy (average recall across classes) has also been used along with accuracy.
- ❖ Results for different methods are shared below-

Method	$\alpha = 0.3$		$\alpha = 0.6$		$\alpha = 0.9$		<i>iid</i>		Average		Real	
	BT	SC	BT	SC	BT	SC	BT	SC	BT	SC	ACC	BACC
Centralized	x	x	x	x	x	x	x	x	70.30	61.01	60.81	47.80
FedAVG	66.24	48.30	66.49	49.15	64.21	54.23	68.52	49.15	66.36	50.21	60.24	50.96
FedProx	66.24	48.30	63.95	51.69	65.23	49.15	70.05	50.84	66.37	49.99	59.52	48.89
MOON	67.25	50.84	63.96	50.00	65.48	53.38	69.29	47.45	66.49	50.42	59.95	50.90
FedFocal	67.00	49.15	65.48	49.15	67.26	54.23	64.21	51.69	65.99	51.05	59.97	49.80
FedCLIP	68.78	53.38	66.24	55.93	67.26	54.23	66.50	50.0	67.19	53.38	58.43	51.17
Ours	82.74	56.78	82.23	58.47	81.73	56.78	82.23	57.62	82.23	57.41	72.37	63.61

- ❖ FACMIC outperforms all the existing methods by quite good margin. The following graph shows the accuracy on the BT data -

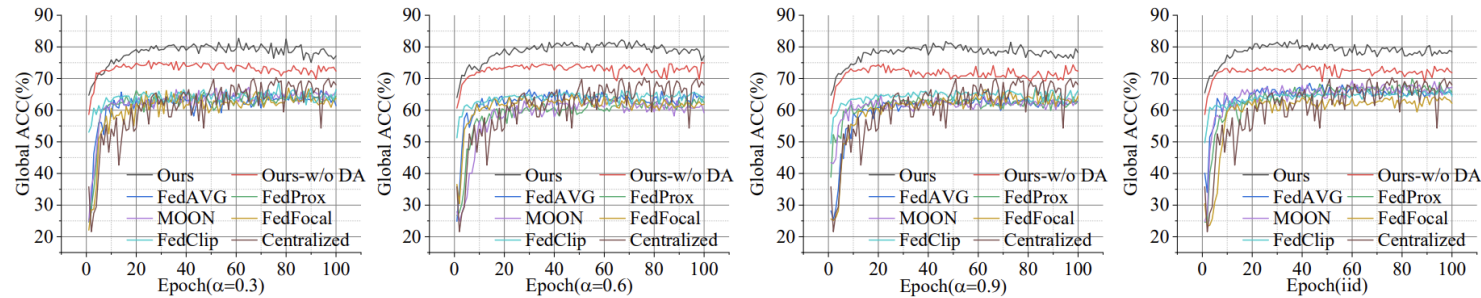


Fig. 2: Global testing accuracy (%) for each round on the BT dataset.

RESULTS

Ablation Study

❖ Effect of $\mathcal{L}_{DA}$:

Adding the domain adaptation loss $\mathcal{L}_{DA}$ leads to a better classification accuracy in the BT dataset, in all situations.

Table 3: Impact of the domain adaptation loss (ACC%).

$\mathcal{L}_{DA}$	$\alpha=0.3$	$\alpha=0.6$	$\alpha=0.9$	<i>iid</i>
✓	82.48	82.23	81.73	82.23
✗	75.63	75.12	74.62	74.62

❖ Effect of Batch size:

The model is robust to batch size on the BT data, however, using a too small batch size for the large-scale real data can result in negative adaptation.

Table 4: Impact of batch size (ACC%).

Batch size	4	8	16	32	Avg
BT (<i>iid</i>)	79.69	80.20	81.22	82.23	80.84
Real	63.39	67.32	69.68	72.37	68.19

❖ Generalization :

To assess FACMIC’s generalization ability, we use the fine-tuned global model trained on the BT dataset under iid condition to perform classification directly on another brain tumor dataset (BT2) without any fine-tuning. The BT2 dataset comprises 7023 samples() has the same classes as the BT dataset. FACMIC here also achieves the best accuracy compared to pre-existing methods.

Table 5: Global testing accuracy (%) using BT2 dataset.

Table 5: Global testing accuracy (%) using BT2 dataset.						
FedAVG	FedProx	MOON	FedFocal	Centralized	FedCLIP	Ours
84.18	84.72	82.32	79.99	86.12	91.7	94.42

CONCLUSION

- By freezing the large CLIP backbone and training only a lightweight attention module, it significantly reduces communication cost.
- The incorporation of class-wise LMMD enables effective domain alignment across heterogeneous clients.
- Experimental results demonstrate strong performance under non-iid and real multi-source settings.

FUTURE WORKS

- Instead of fixed prompt we can use prompt learning or dynamic prompt that can enhance model performance.
- Instead of fixed lambda for calculating our final loss we can use dynamic lambda as our training goes on because strong domain alignment may hurt learning.
- Instead of using CLIP we can use pre-trained BiomedCLIP as the backbone which may increase the performance of the model.

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THANK YOU